

1. Consider a line's function $f(x) = \frac{-9}{3}x + 8$ where any point on the line can be expressed as $(x, f(x))$. A line segment on that line starts at $x = 11$ and ends at $x = -5$. Using the Midpoint Line Drawing Algorithm, **calculate** the first 5 pixels of that line segment.
2. Will the output of the DDA and Midpoint algorithm always be the same for a line segment? If not, how can they be different? Explain with proper reasoning.
3. A red light source with maximum intensity $I_{max} = 1$ has an RGB color of (255, 100, 50) and partially overlaps a 4×4 pixel grid. The overlap area of Pixel A is 20%, Pixel B is 50% and Pixel C is 80%. Calculate the RGB intensity values for each pixel. If the light intensity is doubled ($I_{max} = 2$), how will the final RGB values change? **[When a computed RGB value exceeds 255, we set it to 255 instead of allowing it to go beyond.]**
4. Consider a straight line with endpoints $(-5, 17)$ and $(8, -7)$.
 - a. Using the DDA line drawing algorithm, Determine the first 4 pixels (after the starting pixel) of the line segment (Show the steps of your calculation).
 - b. Using the midpoint line drawing algorithm, compute the first 6 pixels (including the starting pixel) to be colored for the given line segment in their original zone. (Show the steps of your calculation)
 - c. Which line-drawing algorithm would you prefer? Explain your reasoning shortly.

Deadline: 4 March 2025, 11:59PM